# SAURABH SINGH

**Senior Software Engineer · Full-Stack & Distributed Systems · Payments Infrastructure**

hello@iamsaurabh.in | +91 85768 58692 | Bengaluru, India | iamsaurabh.in | saurabhsinghrbz | SaurabhSinghRbz

## SUMMARY

Senior Software Engineer with 4+ years building payments and distributed-systems infrastructure at Pine Labs (Qwikcilver · Woohoo). Own the Go service running the gift-card lifecycle behind HDFC, Axis and Woohoo storefronts, built and ran a 500+ TPS message-queue platform solo for 18 months, and shipped a multi-tenant UAE payments platform end to end.

## PROFESSIONAL EXPERIENCE

**Pine Labs (Qwikcilver · Woohoo)** Senior Software Engineer — Apr 2024 – Present · Bengaluru

Go (chi, GORM) · MySQL read/write split · MongoDB · Redis · Message Queues · OpenTelemetry · AWS ECS

- Own the gift-card order lifecycle in a Go service — activation, shipping, wallet credit, PIN reset, cancel and reissue.
- Architected an async monitor-queue pipeline with row-level idempotency — 100,000+ card bulk orders, no duplicates.
- Kept irreversible card operations safe using explicit state machines, pre-commit audit records and reconciliation jobs.
- Split hot balance and status reads onto MySQL replicas; traced with OpenTelemetry and shipped blue/green to AWS ECS.

**Pine Labs (Qwikcilver · Woohoo)** Software Engineer — Nov 2022 – Mar 2024 · Bengaluru

Node.js · Express · MongoDB · Redis · ActiveMQ (STOMP) · PostgreSQL · Hasura GraphQL · Vue.js · Docker

- Built and ran a multi-tenant B2B message-queue platform solo for 18 months — 500+ TPS, millions of messages daily.
- Guaranteed exactly-once activation on at-least-once delivery via idempotency, dedupe keys, backoff retries and DLQs.
- Migrated the broker from per-call Iron.io to self-managed ActiveMQ behind a pluggable layer, turning linear spend into flat cost.
- Shipped a Vue operations console with RBAC, bulk uploads and per-message tracing and replay, making support self-serve.
- Extended the notification service behind every email, SMS, WhatsApp and OTP — templates, failover, PII encryption.

**Masai School** Full-Stack Development Trainee — Sep 2021 – Nov 2022 · Bengaluru

2,200+ hours of full-time MERN training (1,200 building, 1,000 DSA) · 7 applications shipped · Comeback Kid & Rising Star awards

## SELECTED PROJECTS

**Woohoo** Gift-Card Commerce Platform, Pine Labs — woohoo.in · Live

Go · Node.js · MySQL · MongoDB · PostgreSQL · Redis · ActiveMQ · AWS ECS

- Three services I own form one order path — the Go core queues work, the platform returns idempotent stages, delivery notifies.
- Powers issuance and activation for a 300+ brand catalog across woohoo.in and the HDFC, Axis and FairPrice storefronts.

**Smooth** Payments, POS & Loyalty Platform, UAE — smoothpay.ai · Live

NestJS · Prisma · PostgreSQL · Redis · Socket.IO · Stripe · React + Vite

- Owned a multi-tenant QR-ordering and payments platform end to end — data model, API, dashboards, customer app, deploy.
- Integrated Stripe, Telr and AFS with Apple Pay/Google Pay, multicurrency links, and idempotent row-locked settlement.
- Live order state over Socket.IO, JWT/OAuth tenant isolation, and a rewards and loyalty ledger with guest-claim tokens.

## TECHNICAL SKILLS

**Languages** Go, TypeScript, JavaScript, SQL

**Backend** Node.js, Express, NestJS, chi, GORM, Prisma, Hasura GraphQL, REST APIs, Microservices

**Data** PostgreSQL, MySQL, MongoDB, Redis, ActiveMQ, Query Optimization

**Frontend** React, Next.js, Vue.js, Zustand, Tailwind CSS

**Cloud** AWS (ECS, S3, ECR, KMS), GCP, Docker, nginx, CI/CD, OpenTelemetry, New Relic

**Architecture** Distributed Systems, Event-Driven Architecture, System Design, Idempotency, Multi-Tenancy, JWT/OAuth, RBAC

## EDUCATION & CERTIFICATIONS

**B.C.A., Information Technology** — Amrita Vishwa Vidyapeetham — 2024 – 2026

**Diploma, Mechanical Engineering** — Government Polytechnic, Lucknow — 2019 – 2022

**Certifications** Google Cloud Computing Foundations (2026) · Docker & Kubernetes (2024) · HackerRank JavaScript & REST API

Please visit iamsaurabh.in or github.com/SaurabhSinghRbz for detailed architecture write-ups and more projects.